

Efficiency Just Makes WaterSense

by Cindy Simbanin

Which saves more energy—turning off your faucet or turning off your lights? If you guessed turning off your lights, think again. Running your faucet for 5 minutes can use as much energy as running a 60W light bulb for 14 hours—depending on where your water comes from! Whether it's in the media or around the water cooler, America is talking about the environment. Yet while energy costs and energy efficiency have caught most people's attention, our nation is just starting to focus on water efficiency and its impact on energy use.

Why water efficiency? There are two core reasons, one environmental and one financial. Efficiency makes environmental sense because water is a precious, finite resource that we must take immediate action to protect. Efficiency makes financial sense because even simple water-saving strategies can lead to cost savings for families, businesses, and local and regional economies. It makes financial sense for companies that develop new water-efficient products and technologies, too.

The need to evaluate our use of water supplies is becoming more critical than ever—from the backyard sprinkler to the office cafeteria to the local Laundromat. The U.S. General Accounting Office (GAO) has estimated that, even assuming nondrought conditions, at least 36 states will suffer some sort of water shortage by 2013. Using water efficiently can reduce the impact of these expected shortages. Adopting even a handful of water-efficient practices at home and in business can decrease our total water use by millions of gallons—up to a 30% reduction—each day. And adopting these practices is relatively easy; it can



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Water down the drain means wasted energy also. Running your faucet for 5 minutes can use as much energy as burning a 60W light bulb for 14 hours.

quickly become second nature, just like recycling soda cans or purchasing Energy Star products.

Saving water can mean saving energy for utilities and communities as well. In this country a significant amount of energy is used to treat, pump, and heat water. For drinking water and wastewater systems, the cost of this energy runs to several billion dollars a year, and much of this cost is borne by ratepayers and municipalities. In fact, the cost of the energy used to run drinking water and wastewater systems can represent as much as one-third of a municipality's energy bill.

WATERSENSE MAKES SENSE FOR COMMUNITIES

To promote water efficiency and to help protect the future of our nation's water supply, EPA launched

the WaterSense program in the spring of 2006. WaterSense was created in response to the realization that managing water supplies is an increasingly important issue to communities across the country. In 2002, an EPA report identified a \$224 billion gap between planned funding for new dams, wastewater treatment facilities, and other infrastructure and the actual revenue available to meet identified water supply needs. Demand-side management through increased water efficiency could go a long way to reduce this gap. After all, it is much more cost-effective to find ways to save water than it is to build more dams and more treatment facilities.

With the growing need for more infrastructure, the price of water is on the rise. The WaterSense program's preliminary research indicates that

water and wastewater rates, which are growing 7%–12% annually, are outpacing inflation.

HOW WATERSENSE WORKS

WaterSense has built a national brand for water efficiency, using a symbol that represents the importance of protecting water resources in the United States. Manufacturers can earn the WaterSense label for products that have been independently tested and certified by a licensed certifying body to meet EPA criteria for water efficiency and performance. Generally speaking, WaterSense labeled products will be about 20% more water efficient than the average product in the same category. Testing protocols developed by EPA are used by third-party certification bodies to ensure that WaterSense labeled products perform their intended function as well as—or better than—their less efficient counterparts.

WaterSense also provides technical information to a range of organizations through formal partnership agreements, and recognizes leadership in water efficiency. More than 150 organizations and professionals from across the country in a range of industries have joined the WaterSense program since WaterSense released its first specifications in October 2006. WaterSense partners currently include manufacturers; utilities and water districts; local, state, and federal governments; certified irrigation professionals; trade associations; and professional certifying organizations—all committed to promoting water-efficient techniques.

In addition to certifying products, WaterSense is labeling certification programs for irrigation professionals. These programs assess professional competency in designing, installing and maintaining, or auditing water-efficient landscape irrigation systems. The first two programs to earn the WaterSense label were the Irrigation Association's Certified Irrigation Designer and Certified Irrigation

How Much Water Do We Use?

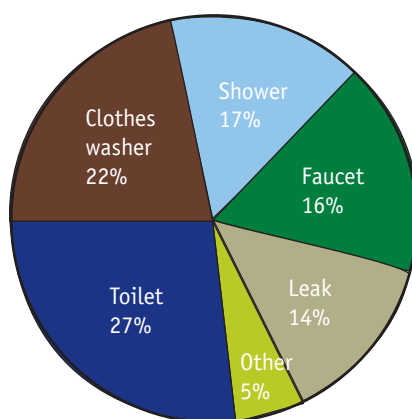


Figure 1. In existing homes, toilets and clothes washers combined make up almost half of all indoor water use. Source: American Water Works Association Research Foundation, "Residential End Uses of Water," 1999.



Contractor programs. WaterSense is also conducting research on water-efficient irrigation technologies. The first outdoor products that will be eligible for labeling will be weather-based irrigation control systems and soil moisture sensors.

The program has also completed specifications for labeling high-efficiency toilets (HETs.) These toilets use less than 1.3 gallons per flush and undergo independent performance testing and certification. Draft specifications for high-efficiency bathroom sink faucets and faucet accessories such as aerators have also been released for public comment.

WATER USE IN NEW HOMES

While water utility managers are aware of the benefits of water efficiency programs, they need more

information on water use patterns in new homes to help develop these programs (see Figure 1 for indoor water use in existing homes). Demand for new homes is on track to total as many as 20 million units between now and 2015, according to a recent Harvard University study. To provide an empirical basis for understanding water use in these homes, EPA has funded a nine-city research study that will collect data from several large water utilities across the United States.

The study will compare standard new homes and high-efficiency new homes that are built to enhance water efficiency. The results of the study will inform the EPA as it works to set targets for builders who wish to provide buyers with more water efficiency options, and to develop specific performance criteria to help consumers identify water-efficient homes. They will also help states and water utilities to establish performance criteria for water use in new homes.

WaterSense will continue to promote water efficiency throughout the country, and to develop specifications for new products and programs. Finally, WaterSense will continue to promote consumer education and outreach aimed at changing the way Americans think about water use in their homes.

Cindy Simbanin is the manager of EPA's WaterSense program.

For more information:

For more information about the program or a list of WaterSense partners, labeled products, or programs, visit www.epa.gov/watersense.

For more information on the draft specification for high-efficiency bathroom sink faucets, go to www.epa.gov/watersense/partners/specs/faucet_background.htm.